CHE-494: Biopharmaceutical Process Development and Manufacturing

Project Assignment

Guidelines:

1. **Please do not share and discuss your assignment with others.**
2. **Two case studies italicized to tailor your answer accordingly.**
3. **Prepare a slide presentation to answer and communicate with the class.**
4. **You may please work on the questions during the course of instructions as and when pertinent classes are delivered. Hence there is no separate homework or written quizzes.**
5. **Questions cover all the course contents.**

<u>Questions:</u>

1. What makes you excited (or not) to work for biopharmaceutical company?
2. *<u>A soluble small molecule drug substance</u> (active pharmaceutical ingredient, API) has <u>very bad physical properties</u>, namely a bimodal particle size distribution with a very small average particle size (less than 20 microns), poor flowability and a very low bulk density. What drug product process would you recommend for solid dosage form? Provide a process flow chart and outline process design considerations for the unit operations involved*.
3. What do you mean by reliable process scale-up? What does it mean in the context of chemical reactions involving reactions of solids with liquid reagent?
4. Why are the physical properties of drug substance important for the drug product process to make solid dosage form?
5. What is the most prevalent unit operation in small molecule drug substance process? Why?
6. What is the most difficult unit operation in the large molecule drug product process? Why?
7. *A small molecule drug substance (active pharmaceutical ingredient, API) has solubility measured in two solvent system (water and ethanol) as a function of temperature and solvent composition as follows.*

Water (wt%)	Temperature (°C)	API Concentration (g/kg solution)
0	25	237.9
0	35	378.4
0	50	648.4
10	25	283.2
10	35	419.1
10	50	689.2
25	25	266.6
25	35	395.3
25	50	679.1
40	25	186.2
40	35	304.8
40	50	570.8
55	25	87.9
55	35	149.5
55	50	377.0
75	25	16.2
75	35	32.9
75	50	101.6

What type of crystallization process (e.g. cooling, antisolvent, etc.) would you recommend to maximize the yield? Outline critical considerations for the reliable scale-up of crystallization process under consideration.

8. Describe key elements of cell-culture process along with the critical considerations? What does the reliable process scale-up mean to you in the context of cell culture process in a bioreactor?